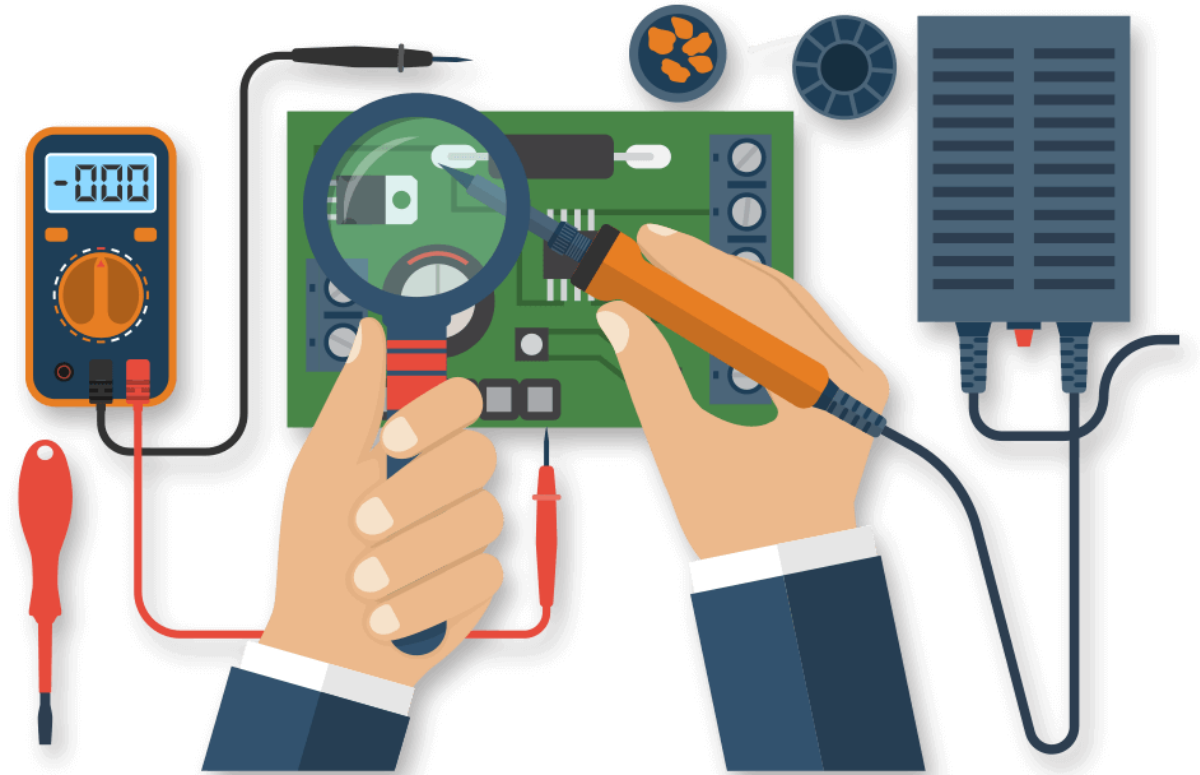


# Computer System Architecture

Day 3: Ohm's Law



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# Outline

- Ohm`s Law
- Electronic Components



# Ohm`s Law

Combining the elements of voltage, current, and resistance, Ohm developed the formula

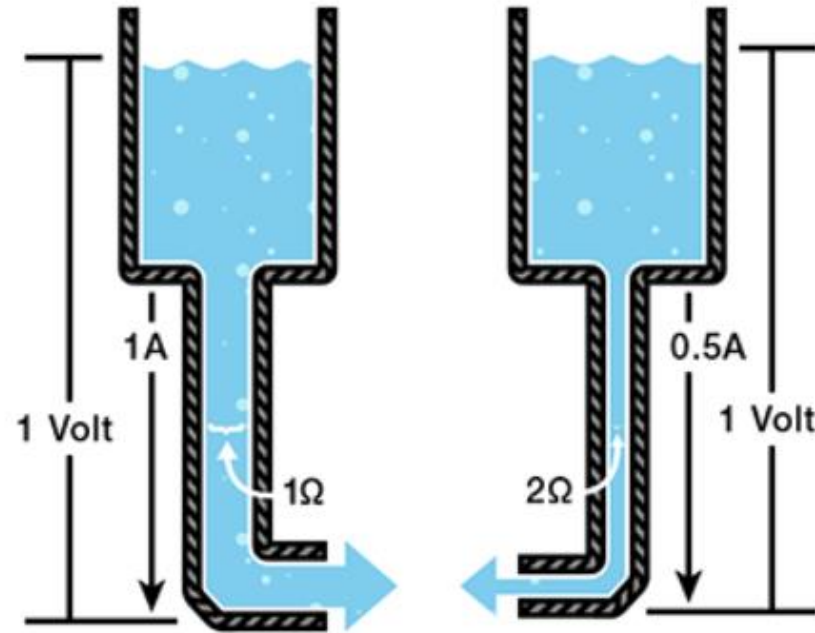
$$V = I \cdot R$$

V = Voltage in volts

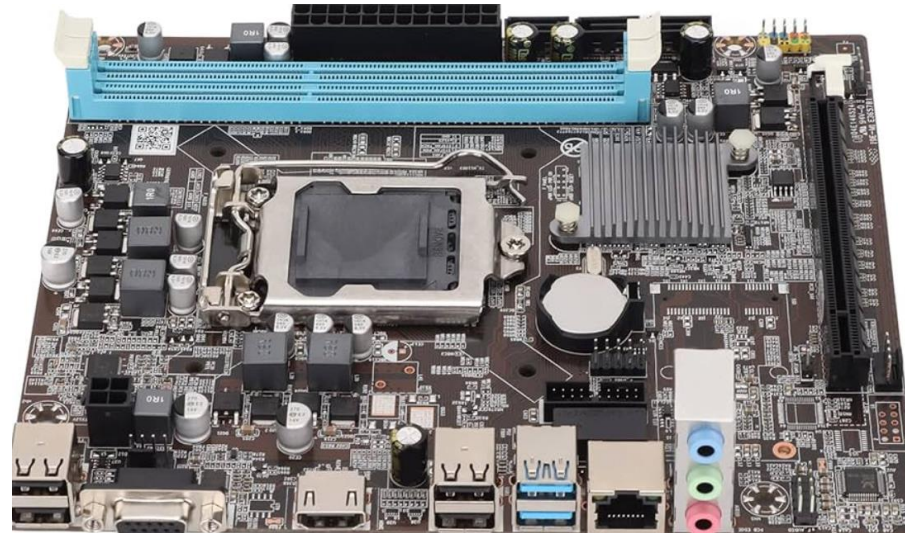
I = Current in amps

R = Resistance in ohms

# Ohm's Law Cont'd...



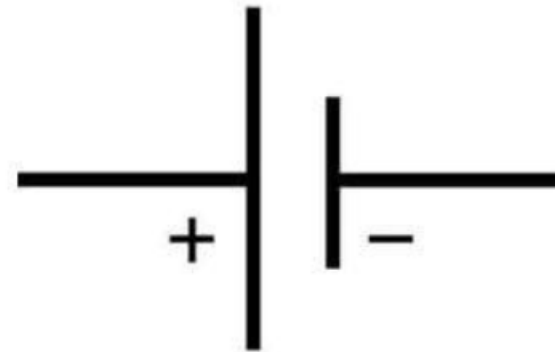
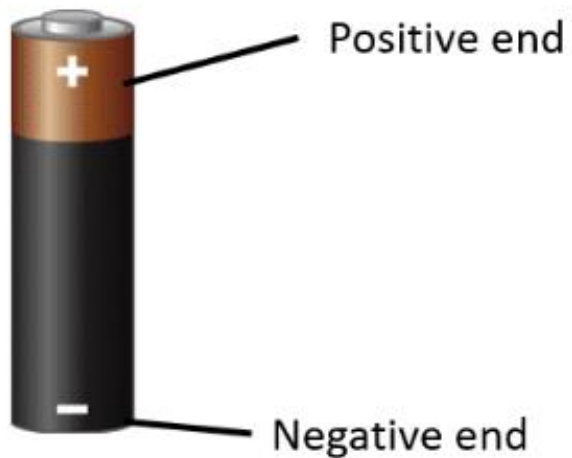
# Basic Electronic Components



# Basic Electronic Components Cont`d...

## Battery

A source of electric power, Provides electric energy to system



Symbol

# Basic Electronic Components Cont`d...

## Resistor

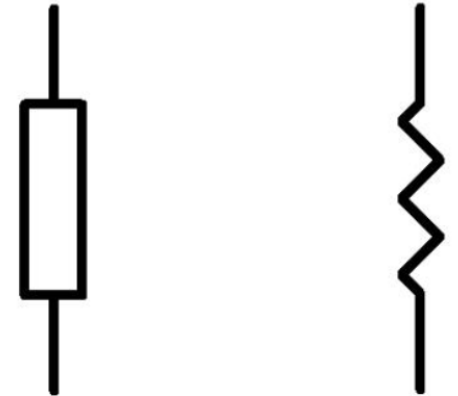
used to oppose or limit the current



Through hole



SMD: Surface Mount Device

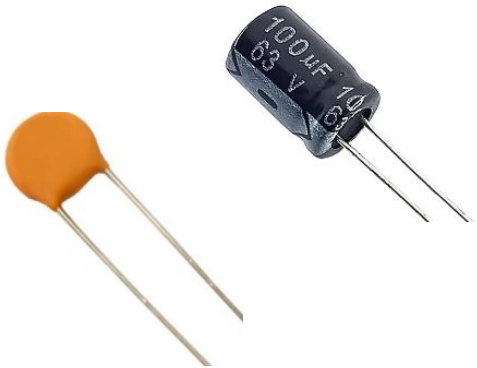


Symbol

# Basic Electronic Components Cont'd...

## Capacitors

Stores energy in the form of electric field



Through hole



SMD: Surface Mount Device



Symbol



# Basic Electronic Components Cont'd...

## Inductors

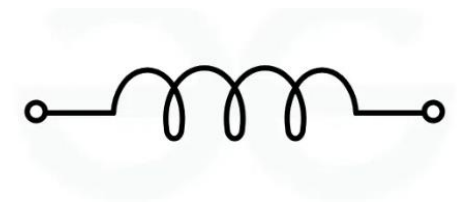
Stores energy in the form of magnetic field



Through hole



SMD: Surface Mount Device



Symbol

# Basic Electronic Components Cont`d...

## Diodes

allows current to flow in one direction



Through hole



SMD: Surface Mount Device



Symbol

# Basic Electronic Components Cont'd...

## Transistors

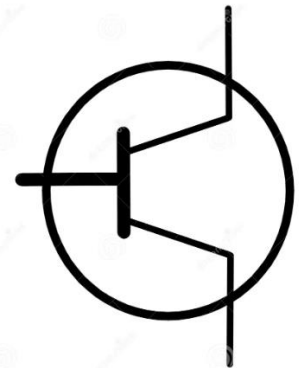
Can be used as a switch or amplifier



Through hole



SMD: Surface Mount Device

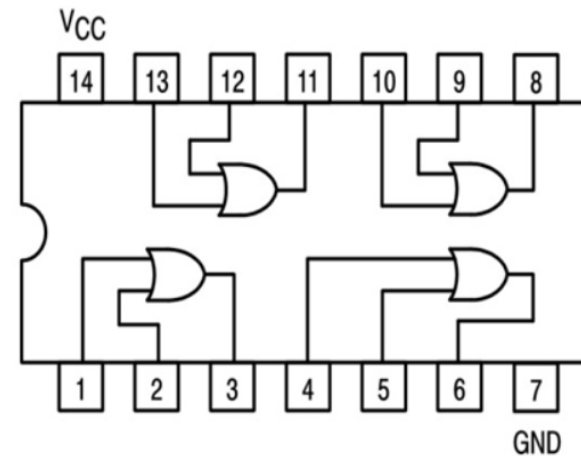
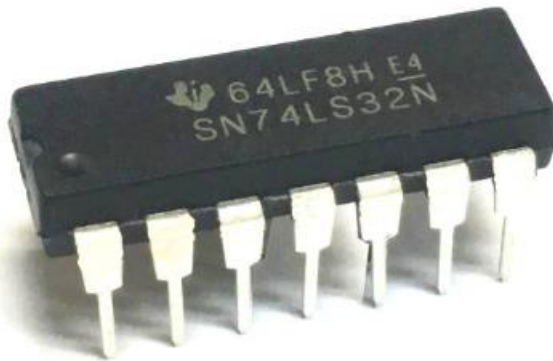


Symbol

# Basic Electronic Components Cont'd...

## ICs

Combines multiple electronic components into a chip



Thank you!



# Questions

